

## Project Planning Phase

# Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| Date          | 29/09/2026                                                          |
| Team ID       | SWTID-2026-7994                                                     |
| Project Name  | AI StudyBuddy - AI assistant for smarter learning and study support |
| Maximum Marks | -                                                                   |

## Product Backlog, Sprint Schedule, and Estimation

| Sprint   | Functional Requirement (Epic) | USN    | User Story / Task                                                                     | Story Points | Priority | Team Members |
|----------|-------------------------------|--------|---------------------------------------------------------------------------------------|--------------|----------|--------------|
| Sprint-1 | Project Setup                 | USN-1  | Create project structure, install dependencies, configure .env and MongoDB connection | 3            | High     |              |
| Sprint-1 | Registration                  | USN-2  | Register with name, email and password (bcryptjs hashing)                             | 2            | High     |              |
| Sprint-1 | Login                         | USN-3  | Login and issue JWT access and refresh tokens                                         | 2            | High     |              |
| Sprint-1 | Access Control                | USN-4  | Auth middleware and admin-only role check                                             | 3            | High     |              |
| Sprint-2 | Study Materials               | USN-5  | Upload material with multer (.txt, .md, .pdf, 5 MB)                                   | 5            | High     |              |
| Sprint-2 | Study Materials               | USN-6  | View and manage own materials                                                         | 3            | Medium   |              |
| Sprint-2 | AI Summary                    | USN-7  | Gemini integration and summary generation                                             | 5            | High     |              |
| Sprint-3 | AI Flashcards                 | USN-8  | Generate and save flashcards                                                          | 3            | High     |              |
| Sprint-3 | AI Quiz                       | USN-9  | Generate and save multiple-choice quiz                                                | 5            | High     |              |
| Sprint-3 | AI Study Plan                 | USN-10 | Generate study plan from goal, hours and days                                         | 5            | High     |              |
| Sprint-4 | History                       | USN-11 | List previously generated resources                                                   | 3            | Medium   |              |
| Sprint-4 | Administration                | USN-12 | Admin user management and usage monitoring                                            | 5            | Medium   |              |
| Sprint-4 | Quality                       | USN-13 | Global error handling, validation and health check                                    | 3            | Medium   |              |
| Sprint-4 | Testing                       | USN-14 | Postman API testing and documentation                                                 | 3            | Medium   |              |

## Project Tracker, Velocity & Burndown Chart

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|------------------------|------------------------------|
| Sprint-1 | 10                 | 6 Days   |                   |                           |                        |                              |
| Sprint-2 | 13                 | 6 Days   |                   |                           |                        |                              |
| Sprint-3 | 13                 | 6 Days   |                   |                           |                        |                              |

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|------------------------|------------------------------|
| Sprint-4 | 14                 | 6 Days   |                   |                           |                        |                              |

Sprint dates and completed points are left blank for the team to fill as sprints run.

### Velocity:

Total planned story points are 50 across 4 sprints, an average of 12.5 points per sprint. With 6-day sprints (24 days), the average velocity is about 2.1 story points per day ( $50 / 24$ ). Recalculate after each sprint using the points actually completed.

### Burndown Chart:

A burndown chart plots remaining work against time. The ideal line falls evenly from the sprint's total story points to zero at the planned end date; the actual line is plotted from the points completed each day. Update it daily from the tracker above.